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Original Research

Enhancing Biomedical Relation Extraction with Transformer Models using Shortest Dependency Path Features and Triplet Information

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ABSTRACT

Entity relation extraction plays an important role in the biomedical, healthcare, and clinical research areas. Recently, pre-trained models based on transformer architectures and their variants have shown remarkable performances in various natural language processing tasks. Most of these variants were based on slight modifications in the architectural components, representation schemes and augmenting data using distant supervision methods. In distantly supervised methods, one of the main challenges is pruning out noisy samples. A similar situation can arise when the training samples are not directly available but need to be constructed from the given dataset. The BioCreative V Chemical Disease Relation (CDR) task provides a dataset that does not explicitly offer mention-level gold annotations and hence replicates the above scenario. Selecting the representative sentences from the given abstract or document text that could convey a potential entity relationship becomes essential. Most of the existing methods in literature propose to either consider the entire text or all the sentences which contain the entity mentions. This could be a computationally expensive and time consuming approach. This paper presents a novel approach to handle such scenarios, specifically in biomedical relation extraction. We propose utilizing the Shortest Dependency Path (SDP) features for constructing data samples by pruning out noisy information and selecting the most representative samples for model learning. We also utilize triplet information in model learning using the biomedical variant of BERT, viz., BioBERT. The problem is represented as a sentence pair classification task using the sentence and the entity-relation pair as input. We analyze the approach on both intra-sentential and inter-sentential relations in the CDR dataset. The proposed approach that utilizes the SDP and triplet features presents promising results, specifically on the inter-sentential relation extraction task. We make the code used for this work publicly available on Github.[1]

1. Introduction

The availability of curated biomedical scientific literature databases and knowledge bases such as PubMed[2] and PubMed Central[3] have enabled numerous research activities aimed at extracting the knowledge stored in the literature and making it available in formats suitable for automated processing. One of the basic steps in making the literature amenable to data mining approaches is the identification of the mention of important domain entities (drugs, diseases, genes, etc.).

A second important step is the detection and extraction of relations between biomedical entities. The manual extraction of such relations and the creation of knowledge bases would be quite time consuming, expensive and inflexible.

The effort to promote different challenges for automatic biomedical relation extraction has led to the availability of annotated corpora, increasing the possibility of training supervised machine learning and deep learning models for automating such tasks. Some of these challenges include the BioCreative V [1] and BioCreative VI [2] competitions, which consist of tasks such as biomedical named entity recognition and biomedical relation extraction. These challenges have released many datasets, among them of particular interest for this work are the Chemical Disease Relation (CDR) and the Chemical Protein

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[1] https://github.com/vanikanjirangat/BioRelation-Extraction_BERT_SDP_Triplets
[2] https://www.ncbi.nlm.nih.gov/pubmed
[3] https://www.ncbi.nlm.nih.gov/pmc/

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Relation (ChemProt) datasets, which are widely utilized in biomedical relation extraction research.

The CDR dataset includes the Chemical Induced Disease (CID) relation extraction task, which covers both intra-sentential and inter-sentential relation extractions with binary relations, while the ChemProt dataset involves only intra-sentential relations with multi-class labeling. These datasets are either annotated at document-level (CDR) or sentence-level (ChemProt). In the CDR dataset, mention level gold annotations are not explicitly specified. This means that the relations between entities are specified only at the document or abstract levels and not at the sentence levels. Hence the sentences that contain an entity pair may or may not represent the relationship between the entities. Thus, the samples that represent positive (a relation exists) and negative relations (no relation) have to be constructed from the given abstract/text.

A relation is defined between two entities, based on whether there exists an association or connection between them, where one entity impacts the other. Specifically, CID relations are defined in the CDR dataset, where chemicals and disease act as entities. According to the task description, a relation exists between these entities if the disease is induced or caused by the given chemical. Similarly, in the ChemProt dataset, a relation exists between the chemical and the protein entity if the protein is affected favorably or adversely by the chemical entity. Due to the inherent challenges in the construction of data samples and the presence of both intra and inter-sentential relations, the CDR dataset is widely employed in literature for biomedical relation extraction studies.

The existing literature proposes multiple strategies for effective sample selection. One of the strategies includes considering the entire abstract for each entity pair whose relation has to be identified. Another popular approach is to consider all the sentences where either a chemical or disease entity mention occur. Thillaisundaram et al. [3] consider the former abstract-level strategy, while Chen et al. [4] propose considering the latter strategy. Another approach was proposed by Verga et al. [5], where the relations between all the entity mention pairs were predicted simultaneously.

These approaches can be time consuming, especially when the text length increases and could also introduce noise to the trained model. For example, consider the sentences related to an entity pair *('neurofibromatosis 1', 'breast cancer')* and the two sentences with the corresponding entity mentions:

1. *Best practice guidelines for* **breast cancer** *detection are not sufficient for the screening of* **neurofibromatosis 1** *carriers.*
2. *An increased* **breast cancer** *incidence and poor survival have been reported for women with* **neurofibromatosis 1**.

Among these two sentences, only the second sentence expresses the actual relation between the two entities, which means here *'breast cancer'* is associated with *'neurofibromatosis 1'*. Hence, the mere presence of the entity mentions in a sentence (sentence 1) does not make it suitable to be considered as a sample that could express the relation. Considering the entire document text, the amount of such noisy sentences could be quite high. The problem could be aggravated with inter-sentence relations, wherein the entity mentions span across sentence boundaries. About 30% of the relations in the CDR dataset are inter-sentential.

Considering these problems, we propose an approach to effectively select the representative samples that express entity relations and perform simple fine-tuning of the pre-trained model. The approach could be closely aligned with the noisy reductions performed in distant supervision methods.

We propose using the Shortest Dependency Path (SDP) features to identify the potential representative sentences for both intra-sentential and inter-sentential relation classifications. The SDP focuses on syntactic structures, which could represent valuable relations between the entities. Recent research has shown the potential of using SDP features [6–9]. We represent the problem as a sentence pair classification task and use the BERT transformer model, specifically the popular biomedical variant of BERT, BioBERT. In representing the sentence pair, we use the sentences selected by the SDP method and the triplet information corresponding to the given sentence as the input pair. The resulting model is compared with different variants to analyze the impact of the proposed components. Further, we also compare our model with the existing state of the art approaches over the CDR dataset.

The rest of the paper is organized as follows. In Section 2, an overview of the existing literature in the field of biomedical relation extraction is presented. Section 3 describes the proposed approach in detail. Data statistics, results, and analysis are discussed in Section 4. Comparisons with state of the art approaches are discussed in Section 5. Finally, conclusions and outlooks are reported in Section 6.

2. Related Work

The datasets released by the BioCreative V and the BioCreative VI challenges have made a wide impact in the biomedical research community, facilitating the use of such annotated data for relation extraction modeling. Various approaches for relation extraction have been presented in the literature, ranging from rule-based approaches and feature-based approaches using Natural Language Processing (NLP) techniques and machine learning techniques to the recent neural models.

In feature-based approaches, syntactic and semantic features were extracted using NLP techniques such as Part of Speech (POS) Tagging, Dependency Parsing, and Chunking for identifying and classifying the entity relations. Dependency parse features such as Shortest Dependency Paths (SDP) information and other tree features were widely utilized to represent the syntactic and semantic relationships between entities [10–14].

Recently, embedding models such as Word2Vec [15], GloVe [16] and their biomedical variants, viz., BioWord2Vec [17] and BioSent2Vec [18], alleviated the tedious task of feature engineering. The advancements of neural models in text classification and retrieval applications, prompted the use of Long Short Term Memory (LSTM) Networks [19] and Convolutional Neural Networks (CNN) [20] in biomedical text mining research [21–24]. Focusing on cross-sentence or inter-sentential relation extraction, Quirk and Poon [25] proposed the use of syntactic parsing while Peng et al. [26] used parse trees encoded by graph LSTM networks.

Another relevant area in literature is the integration of knowledge bases (KB) with neural models. For instance, knowledge guided CNN with SDP features were utilized for chemical induced disease (CID) relation extraction, which achieved about 69.08% F-score [27,28]. Pons et al. [29] and Peng et al. [30] integrated support vector machines (SVM) with knowledge bases and augmented weakly labeled data to obtain F-scores of 70% and 61% respectively.

Recently, Transformer models [31] gained a lot of focus in NLP applications as they facilitated the capturing of informative patterns without any feature engineering or exploiting any external knowledge bases. Bidirectional Encoder Representations from Transformers (BERT) [32] achieved state of the art results in various NLP tasks setting a new hard baseline. The domain specific training of these models improved the performances even more. In the biomedical domain, BioBERT [33] and SciBERT [34] achieved notable performances on biomedical named entity recognition and relation extraction tasks. This motivated the exploration of different variants of these architectures. Wu and He [35] proposed Relation BERT (R-BERT), an entity marking strategy for BERT in relation extraction, which exhibited improved performance in SemEval-2010 Task 8. Saadullah et al. [36] used BioBERT and extended R-BERT for bag-level multi-instance learning while Liu et al. [37] used SciBERT with self-attention structures.

Concerning the strategy of data sample constructions from the CDR dataset, Thillaisundaram et al. [3] proposed to consider the entire text/abstract for each entity pair, while Chen et al. [4] utilized a sentence pair classification approach, with entities and sentences as the inputs. Liu et al. [37] proposed using the sentences in which the target entity pair is located and the sentences between them to construct an instance. Gu et al. [38] presented an approach using sentences with marked entity mentions, together with dependency paths. A novel approach using bi-affine relation attention networks was proposed by Verga et al. [5], which simultaneously predicted all the mention pairs. The authors point out that in the CDR dataset with 500 abstracts, the classification of 5318 candidate mention pairs is required to predict the relation separately for each entity mention pair. This could make the approach quite inefficient. This observation motivated us to explore approaches that could tackle such problems.

This paper proposes a sentence-level approach, where the potential sentence samples that represent an entity relation are selected using the SDP features. In this way, for a particular entity pair, instead of considering all the sentences with an entity mention, a representative sentence is selected which is then used for predicting the relation. The method helps to prune out noisy sentences, which results in improved predictions. We also utilize triplet information in training the pre-trained transformer (BioBert) model with a sentence pair. Our approach is found to achieve comparable performances with the related state of the art approaches that do not utilize any external knowledge sources. The methodology used in the proposed approach is described in detail in Section 3.

3. Methodology

The biomedical relation extraction task can be defined as follows: Given the biomedical abstract $\mathscr{A}$ and biomedical entities $(E_1, E_2, \ldots E_n)$ as input, the task is to identify the relation $\mathscr{R}$ that could exist between any entity pair.

The task is usually formulated as a classification task, where the problem is to classify whether a relationship exists between the given entity pair (E_i, E_j) at sentence-level or document-level. This granularity of whether the classification should be at the sentence or document-levels depends upon the dataset structure, entity annotations and relation annotations. For instance, in the BioCreative VI ChemProt dataset, entities are annotated at mention-level or sentence-level. In contrast, in the BioCreative V CDR dataset, the entities are annotated at the document or abstract levels. Since there are no mention-level annotations in the CDR dataset, we do not know which sentences containing the entity mentions could convey a meaningful relationship. The approach described in subSection 3.1 is proposed for the creation of appropriate training samples from datasets annotated at the document-level.

3.1. Noisy Sample Reductions using Shortest Dependency Path Features

We explain our approach in the context of the CDR dataset for the CID task. Hence, the entity mentions refer to (chemical, disease) pairs.

3.1.1. Constructing Candidate Sentence Sets

For a given (chemical, disease) entity pair, we extract all the sentences from the abstract or document text, which contain entity mentions. If we find the chemical and disease mentions appearing within the same sentence, we consider them a potential intra-sentential mention of the relationship. In the absence of an intra-sentential mention, we check for inter-sentential mentions. In this case, the entity mentions can span across different sentences. Thus, we extract all those sentences where either the chemical or the disease mention occurs. Then we consider a cross-product of all these sentences, which forms the candidate sentence set.

Consider Fig. 1 which represents the annotation of an abstract from the CDR dataset. The entity annotations are represented as *'Chemical'* and *'Disease'*. *CID* indicates the presence of a relationship between the given entity IDs. We represent the abstract as $\mathscr{A}$ and the title as $\mathscr{T}$. The entity annotations $\mathscr{E}$, which include chemical and disease entity mentions, are represented as $\mathscr{C}$ and $\mathscr{D}$ respectively. A relationship between the entities is represented as $\mathscr{R}$. The construction of candidate sentence sets is summarized in the following steps.

1. Initially, we combine the title $\mathscr{T}$, and abstract $\mathscr{A}$, as a single text, let it be the text context, $\mathscr{C}_T$. For each mention of the $(\mathscr{C}, \mathscr{D})$ entity pair, we give a unified representation based on their entity IDs as given in the dataset. For instance, in the given example, the disease mentions, *'systemicsclerosis'* and *'SSc'* corresponds to same disease entity, with entity ID *D012595*. We replace all the occurrences of *'systemicsclerosis'* with *'SSc'*.
2. We split the context $\mathscr{C}_T$ at sentence-level using the *nltk sentence tokenizer*[4].
3. We consider each entity pair and get the sentences within the text context $\mathscr{C}_T$ related to the (C_i, D_j) pair at hand. Let *nc* be the number of chemical entity mentions and *nd* be the number of disease entity mentions. Hence, we have a total of $nc * nd$ (C_i, D_j) pairs.
4. For each (C_i, D_j) pair, we check whether there exists a set of intra-sentential entity mentions. Otherwise, we extract the pair of sentences in which each entity occurs to get all the possible cross-sentence pairs.

This procedure constructs the candidate sentence set, say $\mathscr{S}$, for each entity pair. In the next step, from the candidate sentences, we select the most representative sentence sample or prune out the noisy sentence samples.

We use the Shortest Dependency Path (SDP) features for the sentence selection process in the proposed approach. Making use of the SDP as a feature extracted from the dependency parse tree [39] is one of the most commonly used approaches for extracting the relations between entities. For the proposed approach, we extract two main features from the dependency tree: the shortest-dependency path length and the SDP word sequence. The SDP word sequence is used to extract the triplet information for sentence representation, which will be explained in subSection 3.2.

3.1.2. Selecting Representative Sentence Samples using the Shortest Dependency Paths

In the following paragraphs, we describe the process of extracting the Shortest Dependency Paths (SDP) information for intra-sentential and inter-sentential relation extraction with suitable examples. For a particular chemical-disease entity pair (C_i, D_j), let us assume that S_{intra} is the candidate set of intra-sentential entity mentions. If S_{intra} is empty, we construct the set of inter-sentential entity mentions, which will be denoted as S_{inter}.

Intra-Sentential Relation: The steps involved in extracting SDP features and the path length computations for finding the representative sentences in intra-sentential relations are summarized in the following steps.

1. For each sentence in S_{intra}, we extract the shortest dependency path between the chemical-disease entity mentions, $\mathscr{C}_i$ and $\mathscr{D}_j$ respectively. For constructing the parse tree and extracting the dependency information, we use scispaCy [5], which contains the spaCy models for processing biomedical, scientific or clinical text.
2. Then we compute the path length between $\mathscr{C}_i$ and $\mathscr{D}_j$from the SDP information.
3. Finally we select the sentence from S_{intra} that is associated with the minimum path length.

For instance, consider the entity pair *('cyclosporine', 'thrombotic microangiopathy')* from the example shown in Fig. 1. From Fig. 1, we could find that there is only a single entity mention for each of these

[4] https://www.nltk.org/api/nltk.tokenize.html
[5] https://allenai.github.io/scispacy/

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22836123|t|Late-onset scleroderma renal crisis induced by tacrolimus and prednisolone: a case report.

22836123|a|Scleroderma renal crisis (SRC) is a rare complication of systemic sclerosis (SSc) but can be severe enough to
require temporary or permanent renal replacement therapy. Moderate to high dose corticosteroid use is recognized as a major
risk factor for SRC. Furthermore, there have been reports of thrombotic microangiopathy precipitated by cyclosporine in
patients with SSc. In this article, we report a patient with SRC induced by tacrolimus and corticosteroids. The aim of this
work is to call attention to the risk of tacrolimus use in patients with SSc.

22836123	11	35	scleroderma renal crisis	Disease D007674
22836123	47	57	tacrolimus	Chemical	D016559
22836123	62	74	prednisolone	Chemical	D011239
22836123	91	115	Scleroderma renal crisis	Disease D007674
22836123	117	120	SRC	Disease D007674
22836123	148	166	systemic sclerosis	Disease D012595
22836123	168	171	SSc	Disease D012595
22836123	281	295	corticosteroid	Chemical	D000305
22836123	341	344	SRC	Disease D007674
22836123	386	412	thrombotic microangiopathy	Disease D057049
22836123	429	441	cyclosporine	Chemical	D016572
22836123	459	462	SSc	Disease D012595
22836123	506	509	SRC	Disease D007674
22836123	521	531	tacrolimus	Chemical	D016559
22836123	536	551	corticosteroids Chemical	D000305
22836123	610	620	tacrolimus	Chemical	D016559
22836123	642	645	SSc	Disease D012595
22836123	CID	D016572 D057049
22836123	CID	D000305 D012595
```

Fig. 1. Dataset Annotation Structure from CDR Dataset (from [1]).

chemical and disease entities. Since these mentions co-occur within the same sentence, we have an intra-sentential mention. In this particular scenario, since we have only a single sentence within candidate set S_{intra}, the second and third steps are not applicable. But for the sake of simplicity in illustration, we use this example.

The sentence extracted for the entity pair *('cyclosporine', 'thrombotic microangiopathy')* is given as:

$\mathscr{S}$:*'furthermore, there have been reports of* **thrombotic microangiopathy** *precipitated by* **cyclosporine** *in patients with ssc.'*

Then this sentence is passed into scispaCy to extract the SDP features. The dependency tree obtained from scispaCy is depicted in Fig. 2. The entity mentions are shown in red color. The SDP between the entity mentions is obtained as *['cyclosporine', 'precipitated', 'thrombotic microangiopathy']*. Based on this, the SDP length is 2. The same steps will apply to other sentences if $|S_{intra}| > 1$. In this case, we use the third and final step, where the sentence related to the minimum SDP length gets selected. Note that we use the raw text and the entity pairs for SDP computations without any pre-processing.

In the following paragraph, we define the procedure of extracting the SDP information and computing the path lengths for an inter-sentential entity mention.

Inter-Sentential Relation: To explain the steps involved in the SDP computations and noise reductions for inter-sentential entity mentions, we consider an example from Fig. 1. For instance, we consider the chemical-disease entity pair *('corticosteroid', 'ssc')*. Observing Fig. 1, we could find that there is no sentence where the mentions of these two entities co-occur. Remember that in pre-processing all the entity mentions are replaced with their unique mentions. Hence *'systemic sclerosis'* is replaced with *'ssc'*. Since, $|S_{intra}| = 0$, we extract the sentences where either one of the chemical or disease entity mentions occur and then take a cross-product of all these sentences to get all the possible sentence pairs related to the entity pair mentions.

Let us assume that the candidate set is S_{inter}, which constitutes pairs of sentences, each containing a mention of the chemical entity and disease entity, respectively. We represent S_{inter} with sentence pairs (*Sc*, *Sd*), where *Sc* is the sentence that has the chemical mention and *Sd* the sentence with disease mention.

Considering the above example, we extract 2 sentences with the chemical entity mention, i.e., *'corticosteroid'* and 3 sentences with the disease mention, i.e., *'ssc'*. This creates a total of 6 candidate sentence pairs. The sentences extracted for *Sc* and *Sd* are shown below:

Sc: *['moderate to high dose* **corticosteroid** *use is recognized as a major risk factor for src.', 'in this article, we report a patient with src induced by tacrolimus and* **corticosteroids**'*]*

Sd: *['the aim of this work is to call attention to the risk of tacrolimus use in patients with* **ssc**.*', 'src (src) is a rare complication of* **ssc** *(ssc) but can be severe enough to require temporary or permanent renal replacement therapy.', 'furthermore, there have been reports of thrombotic microangiopathy precipitated by cyclosporine in patients with* **ssc**.*']*

As noted, we can have multiple mentions of the same entity in a sentence. In this case, we consider the first mention. In the next step, we extract the dependency information from among these sentences for SDP length computation. Unlike intra-sentential mention, the SDP computation of cross-sentence entity mentions is not direct. Hence, we employ a strategy described by Gupta et al. [40] for inter-sentential SDP computation. The approach is summarized in the following steps.

1. Consider a sentence pair (say at *ith* and *jth* positions from *Sc* and *Sd* respectively) obtained by the cross-product relation *Sc* x *Sd*. Let the

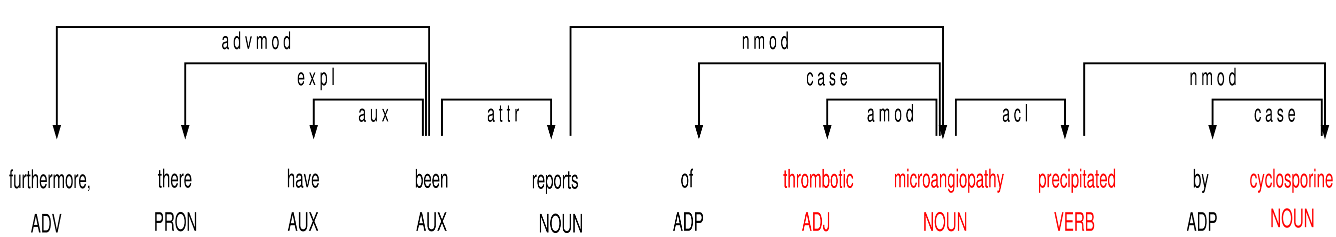

Fig. 2. The SDP between entities 'cyclosporine' and 'thrombotic microangiopathy' for the intra-sentential relation is highlighted, where the path length is 2.

sentences in the pair be Sc_i and Sd_j, such that $Sc_i \in Sc$ and $Sd_j \in Sd$ respectively.
2. Find the *ROOT* word from the dependency tree for sentence Sc_i and compute the SDP of the chemical entity mention from the *ROOT* word.
3. Next, we find the *ROOT* word for sentence, Sd_j and compute the SDP of the disease entity mention from the *ROOT* word.
4. Add the SDP's obtained from 2 and 3 to obtain the final path length between the entity mentions.

Since the illustration with each pair of sentences (6 pairs) would take too much space, we consider one of the sentence pairs for explaining the steps discussed above. Among the sentence pairs constructed from the example above by performing the cartesian product *Sc* X *Sd*, we select the following sentence pair,

Sc_i: *'moderate to high dose* **corticosteroid** *use is recognized as a major risk factor for src.'*

Sd_j: *'src (src) is a rare complication of* **ssc** *(ssc) but can be severe enough to require temporary or permanent renal replacement therapy.'*

We chose this sentence pair because this turned out to be the final representative sentence selected by the proposed approach. Note that our approach considers the first occurrence of the entity mention in the sentence. The procedure is explained with the above example sentence pair in the following steps:

1. Initially, get the dependency parse trees for sentences Sc_i and Sd_j. These are depicted in Figs. 3 and 4 respectively.
2. Secondly, identify or extract the *ROOT* node for sentences Sc_i from the dependency tree. Here, $ROOT(Sc_i)$ = *'recognized'*. Then compute the SDP length from chemical entity mention *'corticosteroid'* and the *ROOT* node. The SDP path terms are marked in red in Fig. 3 which constitutes, *['corticosteroid', 'use', 'moderate', 'recognized']* and hence the SDP length is 3.
3. Extract the *ROOT* node for sentences Sd_j, where $ROOT(Sd_j)$ = *'complication'*. The SDP between disease entity mention *ssc* and the *ROOT* node is shown in Fig. 4 which constitutes, *['ssc', 'complication']* and hence the SDP length is 1.
4. Compute the final path length, which will be the sum of the SDP lengths obtained from steps 2 and 3, which is 4. Hence, the SDP length for the inter-sentential entity pair *('corticosteroid', 'ssc')* is computed as 4 by the proposed approach.

The same steps are repeated for all the sentence pairs obtained by *Sc* X *Sd*. The SDP lengths computed from *ROOT* nodes for the two sentences in *Sc* is *[3,5]*. Respectively for *Sd* is *[6,1,7]*. Hence the SDP lengths for the 6 pair of sentences is *[9,4,10,11,6,12]*. From among these, we find the minimum SDP length and the associated sentence pair. Thus, we identify the representative sentence sample as *('moderate to high dose* **corticosteroid** *use is recognized as a major risk factor for src.','src (src) is a rare complication of* **ssc** *(ssc) but can be severe enough to require temporary or permanent renal replacement therapy.')* with the path length 4. Once the sentence samples are selected, we train the model using the approaches described in Section 3.2.

3.2. Fine-Tuning with the Pre-trained Transformer Model

We use the pre-trained biomedical variant of the BERT model, BioBERT [33], as it was additionally trained on biomedical text from PubMed and PMC and has shown improved performances on biomedical NER and relation extraction tasks. We model the problem of relation extraction as a sentence pair classification task. The paired inputs include the selected sentence with its triplet feature. A triplet, $\mathscr{T}_p$ is represented as (h,r,t), where *h* is the head entity, *t* is the tail entity and *r* is the relation. In the given problem, the entity pair constitutes the head and tail entities, and *r* is represented by the *ROOT* word extracted from the dependency tree of the selected sentence. For instance, consider the sentence *S* and entity pair (C_i, D_j) :

S: *'The aim of the present study was to investigate the protective effect of the Solidago virgaurea extract on* **isoproterenol** *induced* **cardiotoxicity** *in rats.'*, (C_i, D_j) : *('isoproterenol','cardiotoxicity')*

The triplet extracted from *S* is given as T_p, with *'induced'* as the *ROOT* word, T_p: *('isoproterenol','induced','cardiotoxicity')*

Finally, the sentence-pair is represented as,(*S*: *'The aim of the present study was to investigate the protective effect of the Solidago virgaurea extract on* **[isoproterenol] chemical** *induced* **[cardiotoxicity] disease** *in rats.'*, T_p: *('isoproterenol_***induced***', '***induced***_cardiotoxicity')*), where the entity mentions in the sentence are marked.

For inter-sentential relations, we extract the *ROOT* words for each sentence containing one of the entity pair mentions. Hence considering the entity pair as (C_i, D_j) and *ROOT* words as (R_c, R_d), the final T_p is represented as:

$T_p : (C_i\_R_c, D_j\_R_d)$.

The next section reports our experimental results, with a detailed analysis and comparisons of the proposed approach and its variants.

4. Data Statistics, Experimental Setup and Result Analysis

In this section, we briefly discuss the datasets used for the experimentation. We used two datasets: the BioCreative V Chemical Disease Relation (CDR) dataset[6] and the BioCreative VI ChemProt challenge dataset[7]. The focus is mainly given to the CDR dataset for the chemical-induced disease (CID) relation extraction task, which aims to identify the relations between chemical-disease entity pairs. This dataset includes entity mentions at both the intra-sentential level and the inter-sentential level. Further, as discussed, there are no mention-level annotations in the CDR dataset, which allows us to test our proposed approach using SDP for noisy sample reductions.

The CDR dataset, as shown in Fig. 1, provides all the entities present in the abstract and the entities that hold a relation (class 1/positive). All the possible entity pairs excluding the positives fall into the negative category/class 0. We also report the experimental results on the ChemProt dataset, where the task is to find chemical-protein relations. In this dataset, the task is to do a multi-class relation classification, where the positive relations can fall into 5 categories $[CPR:3, CPR:4, CPR:5, CPR:6, CPR:9]$ (CPR: Chemical Protein Relation) and we represent the negative class as $[NOREL]$[8]. The data statistics of both the datasets are given in Table 1.

We used BioBERT-Base v1.1 (+ PubMed 1 M), which refers to the BioBERT model trained on PubMed for 1 M steps as the pre-trained model. The experiments were done using PyTorch Hugging face implementations [9] by fine-tuning the model on the respective datasets. The model is fine-tuned for 10 epochs, using Adam optimizer and a learning rate of 2e-5 on the training data. In the subsequent sections, we discuss the experimental results, analysis, and comparisons in detail.

4.1. Experimental Results using the Proposed Approach in the CDR and ChemProt Datasets

This section presents the results of the proposed approach in the CDR and the ChemProt datasets. The proposed approach refers to the combination of using SDP and triplet features. In the CDR dataset, the data is binary classified, so we have either Class 1 (relation exists/positive) or Class 0 (no relation/ negative), while in the ChemProt dataset, we have

[6] https://biocreative.bioinformatics.udel.edu/tasks/biocreative-v/track-3-cdr/

[7] https://biocreative.bioinformatics.udel.edu/tasks/biocreative-vi/track-5/

[8] Notice that the CPR dataset defines ten categories of relations, but only five of them were effectively used in the shared task. In our work, we make use of the same five categories that were used in the shared task.

[9] https://github.com/huggingface/transformers

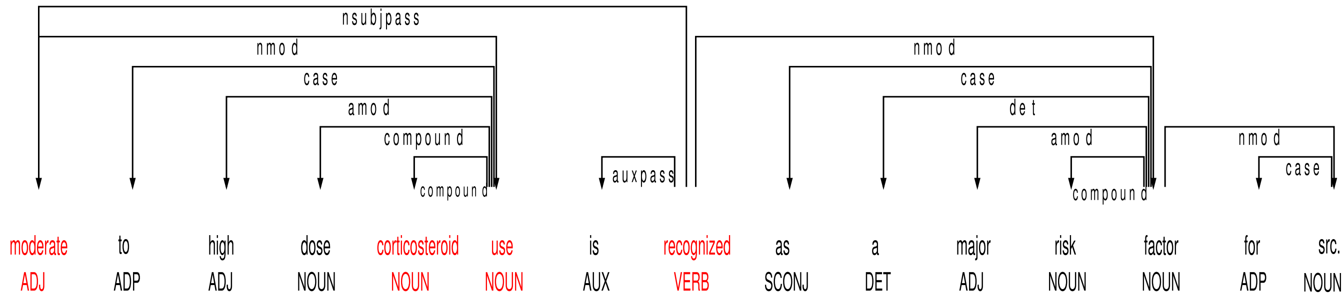

Fig. 3. The SDP between entities ‘corticosteroid’and *ROOT* node ‘recognized’.

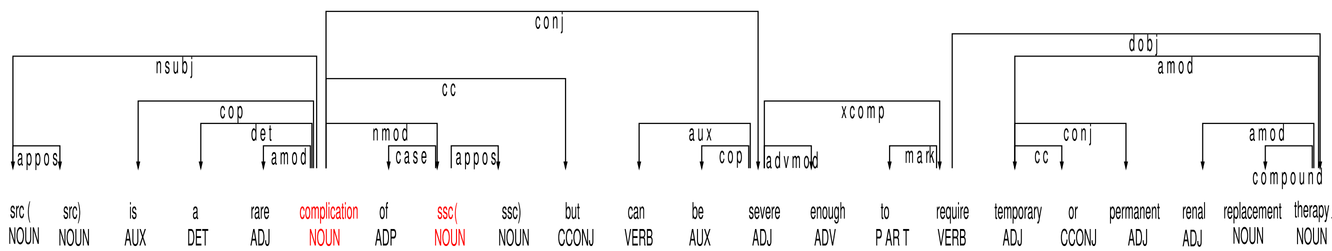

Fig. 4. The SDP between entities ‘ssc’ and *ROOT* node ‘complication’.

Table 1
Data Statistics for CDR and ChemProt Datasets.

Dataset	Training set	Development set	Test set
CDR			
Documents	500	500	500
Positive	1038	1012	1066
Negative	4324	4134	4374
ChemProt			
Total Sentences	6437	3558	5744
Positive	4172	2427	3469
Negative	2265	1131	2275
Classes			
0 (CPR:3)	777	552	667
1 (CPR:4)	2260	1103	1667
2 (CPR:5)	173	116	198
3 (CPR:6)	235	199	293
4 (CPR:9)	727	457	644
5 (NOREL)	2265	1131	2275

5 classes with relation and one negative class (no relation). For the final CDR model, we used both training and development or validation set for model learning.

It could be observed that on the CDR dataset, our approach presented 65% F-score on the positive class. On the ChemProt dataset, we obtain a 74.86% micro-averaged F-score over the positive classes. The ChemProt results are comparable with the BioBert base results. In the next subsections, we present the experimental analysis on different components and variations of the proposed approach over the CDR dataset to understand the contribution of different components/features. (See Table 2).

Table 2
Experimental Results with the Chemical-induced Disease Relation (CDR) and ChemProt Datasets using Proposed Approach.

CDR	Precision	Recall	F1-score
Class 0 (negative)	0.91	0.92	0.91
Class 1 (positive)	0.65	0.64	0.65
ChemProt	**Precision**	**Recall**	**F1-score**
Class 0 (CPR:3)	0.75	0.65	0.69
Class 1 (CPR:4)	0.77	0.81	0.79
Class 2 (CPR:5)	0.76	0.75	0.75
Class 3 (CPR:6)	0.71	0.76	0.73
Class 4 (CPR:9)	0.67	0.69	0.68
Class 5 (NOREL)	0.88	0.88	0.88

4.2. Analyzing Different Components in the Proposed Approach

The experimental analyses were done under two settings as given below:

- Using training data alone (TRAIN ONLY)
- Using both training and development sets (TRAIN AND DEV)

Under both settings, we tested different variations of the proposed approach based on the two main components, viz., Shortest Dependency Path features (SDP) and triplet features. The variations experimented are listed below:

1. **Without SDP_With Triplet (var1)**: Variation 1 does not use the SDP method to select the training samples but uses triplet information in model learning. This variation hence uses a random selection of samples. In this case, we randomly select the sentences from the candidate set.
2. **With SDP_Without Triplet (var2)**: Variation 2 is the approach without triplet information. Here we use the SDP information to select the samples but omit triplets for model learning. In this case, in contrast to the approach described in Section 3.2, we use the entity mentions directly for the sentence-pair classification. Hence, in this variation, instead of T_p: *(‘isoproterenol_induced’, ‘induced_cardiotoxicity’)*, we will have T_p: *(‘isoproterenol’, ‘cardiotoxicity’)*.
3. **Without SDP_Without Triplet(var3)**: In this variation, we analyze the performance by removing both triplet and SDP information.
4. **With SDP_With Triplet (Proposed Approach)**: Finally, we have the variation with both SDP and triplet information, which is our proposed approach.

In Fig. 5, we analyze the impact of different components involved in the proposed approach. From Fig. 5, considering the TRAIN ONLY setting, we could observe that Variation 3, *Without SDP_Without Triplet (var3)*, presents 55% F-score. Using the second variation, *With SDP_Without Triplet(var2)*, an improvement of 5 points is noted (i.e.,

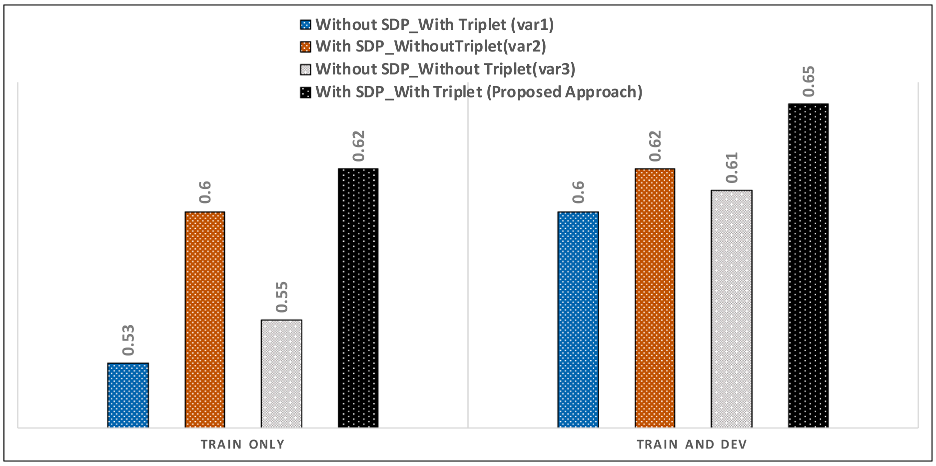

Fig. 5. Comparing the Different Variations of the Proposed Approach.

60% Fscore). This indicates the impact of using SDP information in the proposed approach, which helps in reducing the noisy samples and improving the model performance. But Variation 1, with only triplet information, i.e. *Without SDP_With Triplet (var1)* is found to deteriorate the model performance by reducing the F-score to 53%. Instead, the final model, *With SDP_With Triplet (Proposed Approach)*, boosts the performance to 62%.

It is also interesting to note that the triplet information is effective only in combination with the SDP feature (comparing var1, var2, and var3). In other words, when the samples are randomly selected (without SDP), training the model with triplet feature acts like noisy information and hence degrades the performance. Considering the TRAIN AND DEV settings, similar observations can be made, with the difference that variation 2 with SDP presents only *1* point improvement over Variation 3 without SDP. But the final model with both triplet and SDP information presents a *4* point improvement over *var3*. This indicates that using the triplet feature along with the SDP impacts the performance positively.

Further, we also investigated using the distance between the sentences based on their offset positions to select the inter-sentential samples. In this case, we computed the distance between the selected pairs of inter-sentential samples for an entity pair. This distance is computed as the difference in the sentence offset position. We call this new feature *sentence_distance* feature. This distance indicates the closeness of two sentences in terms of their positions in a document text. The general notion could be that the sentence pairs that lie close to each other may express more meaningful entity relations, except for long-range dependencies. With this assumption, the sentence pair which has minimum *sentence_distance* is selected as the representative sample.

Thus as a final variation, we tested the proposed approach by replacing SDP in inter-sentential relations with the *sentence_distance* feature. For intra-sentential relations, we still used SDP features. We obtained 54% F-score with TRAIN ONLY and 60% F-score for TRAIN AND DEV with the above settings. The performance drops compared to the proposed approach. Another problem with this model was that the presence of multiple sentence pairs with the same *sentence_distance* value. In these cases, we had to take all the sentence pairs into account and take a majority voting.

In the next subsection, we analyze the performances separately on intra and inter-sentential levels.

4.3. Analyzing Intra-Sentential and Inter-Sentential Results (CDR)

This subsection reports the results with intra-sentential and inter-sentential relations separately. We analyze these results since the inter-sentential relation extractions are considered to be more difficult to tackle.

From Table 3, it can be observed that with the proposed approach we obtain 70% F-score on intra-sentential relation classification and 52% F-score on inter-sentential relations. Obviously, the relation classification in the inter-sentential case is much harder than intra-sentential entity mentions. The experimental results obtained on intra-sentential and inter-sentential relations without SDP Information are also reported. With the inter-sentential relations, the F-score increases from 38% to 52% using the proposed sample selections with SDP. This indicates a *14* point F-score improvement on inter-sentential relation. A performance increase of *3* points is noted in intra-sentential relations. This indicates that the proposed approach effectively improves the relation classification, specifically with the inter-sentential relation classifications.

Further, we have also reported the performance with the inter-sentential distance feature. It can be seen that the proposed approach outperforms this model, indicating that the SDP feature is more effective in the selection of representative samples compared to the distance feature. This could also be because the distance feature may not be so prominent as we consider only abstracts and not the full texts in the CDR dataset. The investigations to explore the impacts on longer text or full scientific articles will be carried out in the future.

In Section 5, we present the comparison of performances of the

Table 3
Experimental Results on Intra-sentential and Inter-sentential Relation Classification using Proposed Approach.

Intra-Sentential Relation (Proposed Approach)	Precision	Recall	F1-score
Class 0 (negative)	0.87	0.86	0.86
Class 1 (positive)	0.69	0.71	**0.70**
Intra-sentential Relation (Without SDP_Without Triplet)	**Precision**	**Recall**	**F1-score**
Class 0 (negative)	0.85	0.86	0.86
Class 1 (positive)	0.68	0.66	0.67
Inter-Sentential Relation (Proposed Approach)	**Precision**	**Recall**	**F1-score**
Class 0 (negative)	0.94	0.95	0.94
Class 1 (positive)	0.56	0.48	**0.52**
Inter-sentential Relation (Without SDP_Without Triplet)	**Precision**	**Recall**	**F1-score**
Class 0 (negative)	0.92	0.95	0.94
Class 1 (positive)	0.46	0.32	0.38
Inter-sentential Relation with Distance Feature	**Precision**	**Recall**	**F1-score**
Class 0 (negative)	0.92	0.97	0.94
Class 1 (positive)	0.57	0.30	0.39

proposed approach with the state of the art approaches in biomedical relation extraction on the CDR dataset.

5. Comparisons with the Baselines

This section compares the proposed approach with seven state of the art approaches over the CDR dataset. The results are reported in Table 4, where we compare our approach only to those approaches which do not use any external knowledge bases. Gu et al. [38] employed a Maximum Entropy (ME) model and a convolutional neural network (CNN) for relation extraction. Specifically, they used a Maximum Entropy classifier for inter-sentential relations and a CNN with dependency path features for intra-sentential relations, along with heuristic rules at the post-processing stage. Zhou et al. [41] proposed a feature-based model for lexical features, a tree kernel-based method for syntactic information, and LSTM for semantic information and used the properties of all these models. Li et al. [42] used a recurrent piecewise CNN for relation classification, while Verga et al. [5] proposed a bi-affine network for simultaneously predicting relations for multiple entity mentions. Sahu et al. [43] proposed using graph-CNN for document-level dependency and Liu et al. [37] used SciBERT with self-attention structures.

From Table 4, it can be observed that the proposed approach presents good performance compared with the baselines. Comparing the proposed approach results under the TRAIN + DEV setting with Liu et al. [37], which also utilizes both training and development data, a performance improvement is noted. Comparing with Verga et al. [5], which uses only training data, the proposed approach (TRAIN) presents a comparable performance.

A recent approach proposed by Chen et al. [4] employed a CNN with BERT to classify the relations. In their approach, the constructed test dataset statistics varied from the actual CDR statistics since they generated only 1 negative sample per positive sample. This strategy generated a test set with only 1018 negative instances, while according to the actual CDR dataset statistics it was 4374. They report a 0.7 F-score with 0.71 recall and 0.69 precision. Liu et al. [37] also reports that their constructed CDR dataset had 1042 positive samples and 4376 negative samples, which varied a little from the original one as reported in Table 1.

Our approach generated 4263 negative samples and 1050 positive samples, which is 111 fewer samples in the negative class and 16 fewer samples in the positive class compared to actual CDR statistics as given in Table 1. In the worst case, assuming that all these 111 + 16 missing samples were incorrectly classified, the F-score would drop to 0.618. The results are still comparable with the state of the art approaches.

In Table 5, we compare our results with the two state of the art approaches that report intra-sentential and inter-sentential results separately. Christopoulou et al. [44] used edge-oriented graphs with pre-trained PubMed embeddings for relation extraction. The proposed approach produces a slightly higher F-score value (improved by 1.1 points) compared to the performance of Christopoulou et al. [44]. Hence, based on the reported analysis, we could note that both SDP and triplet features impact the model performance positively.

On a detailed analysis of the predictions of our model, we found some interesting scenarios. For example, consider the instance with the entity pair *('[tacrolimus]chemical', '[SRC]disease')* and the selected sentence:

"Late-onset [SRC]disease induced by [tacrolimus]chemical and prednisolone: a case report.".

The true label of the entity relation was 0, which means there is no relation between the given entities. Our model predicted this as 1. By analyzing the sentence, we could observe that it seems confusing and could be easily misinterpreted by the model, especially with the presence of the word 'induced' between the chemical and disease entities. Another instance was with the entity pair *('[prednisolone]chemical', '[SRC]disease')* on the same sentence.

Besides, the abstract is only a small component of the whole paper, and it might not include all the relevant interactions. Further, when the relation extraction approach is applied in a real situation instead of an experimental one, it is reasonable to expect that relations are extracted from the whole document, rather than just the abstract. Hence, considering the entire document might even improve the performance, which should be investigated.

Table 4
Performance Comparisons with State of the Art Methods on Positive Class (Class 1).

Model/System	Precision	Recall	F1-score
Gu et al.[38]	0.557	0.681	0.613
Zhou et al.[41]	0.556	0.684	0.613
Li et al. [42]	0.552	0.636	0.591
Verga et al.[5]	0.556	0.708	0.62
Sunil e al.[43]	0.528	0.66	0.586
Liu et al.[37]	0.655	0.626	0.64
Proposed approach (TRAIN + DEV)	0.65	0.64	0.65
Proposed approach (TRAIN)	0.65	0.58	0.62

Table 5
Performance Comparisons with State of the Art Methods for Inter-Sentential Relation Extractions on Positive Class (Class 1).

Model/System	Precision	Recall	F1-score
Gu et al.[38]	0.519	0.07	0.117
Christopoulou et al.[44]	0.56	0.467	0.509
Proposed approach	0.56	0.48	0.52

6. Conclusion

This paper proposed an approach that uses the shortest-dependency information to reduce noisy samples and select the most effective representative sentences for training. The approach is found to positively impact the performance in extracting biomedical relations, specifically those where the entities relations are cross-sentential. We also proposed the use of triplet information in model learning, and the experimental analyses have shown the impact of the combinations of SDP and triplet in improving the relation classifications. Further, the proposed approach is simpler but effective in tackling intra and inter-sentential relation extractions with improved or comparable performances compared to the previous approaches.

The selection at the sentence-level may still hamper the overall meaning of some relations. In such cases, the selection at the document-level could be helpful, but at the same time, this could add noisy information. Thus, further investigation of such aspects is important to be considered in sample selection.

In the future, we plan to investigate approaches that could tackle this trade-off between the sentence-level and the document-level approaches. We also want to analyze how well the SDP approach could be leveraged for noisy sample reductions when collecting data for scarce-resource domains, especially for distant supervision methods. We plan to focus more on the challenging inter-sentential relation extraction using graphical models with pre-trained model capabilities, which could facilitate capturing long-distant relations. Another important aspect to consider specifically when dealing with entire documents would be coreference resolutions to identify some relations accurately. We are exploring suitable datasets to extend our experiments to relation extractions for long documents/scientific papers. We plan to focus initially on general datasets such as DocRed [45], and SciREX [46].

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Pseudo-Code

```
FOR each abstract 𝒜 in dataset 𝒟
  Split 𝒜 into sentences to form sentence set 𝒮
  Add the title 𝒯 to the sentence set 𝒮 to form the context 𝒞_T
END FOR
FOR each (Chemical, Disease), (𝒞, 𝒟) pair
  Unify the names based on entity ID's given in the dataset 𝒟
  Extract the sentences corresponding to each (C, D) pair
  Extract the intra-sentential(S_intra) and inter-sentential sentences (S_inter)
  IF S_intra
    COMPUTE SDP between the given (C, D) pair
  ELSE
    FOR each sentence pair (Sc, Sd) in S_inter
      COMPUTE the root nodes for each sentence (Sc, Sd)
      COMPUTE the SDP between (Sc, Sd) and it's root
      COMPUTE the sum of the SDP's
    END FOR
  END IF
  DETERMINE the sentence with minimum SDP, S_m
END FOR
FOR each (S_m, (C, D)) pair
  COMPUTE the triplets T_p
END FOR
FOR each (S_m, T_p)
  COMPUTE the embeddings using BioBERT
END FOR
Train a sentence-pair classification task on top of the BioBERT encoder On testing, predict the relation class
```

1. Time complexity: The sample selection using SDP takes $O(n)$ time, where n is the number of input sentences. BERT sequence classification task has a time complexity of $O(n^2)$ due to the matrix multiplications involved in the attention mechanism. Thus, our model will have a time complexity of $O(n^2)$.
2. Execution time: The model training and inference were done on COLAB https://colab.research.google.com/, with 1 Tesla K80 GPU. The training took approx 40 min, and inference took about 10 min. The initial data selection with SDP took approx 3 min.

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